

* Bernoulli Differential Equations;

(1)

In this section we are going to take a look at D.E in the form:

$$(1) \rightarrow \boxed{y' + P(x)y = q(x)y^n}, n \in \mathbb{R}, n \neq 0, n \neq 1; \text{why?}$$

Case (1): If $\boxed{n=0}$, then equ(1) becomes:

$$\boxed{y' + P(x)y = q(x)} \text{ (First order non-homog.)}$$

Can be solved by Integrating factor method.

Case (2): If $\boxed{n=1}$, then equ(1) becomes:

$$y' + P(x)y = q(x)y \Rightarrow y' = (q(x) - P(x))y$$

$$\boxed{\frac{dy}{y} = (q(x) - P(x))dx} \text{ (separable)}.$$

Case (3) $n \neq 0, n \neq 1$. May be $n = \frac{1}{2}, n = -1, \dots$

Methodology: To solve equ(1):

step (1): Let $\boxed{v = y^{1-n}} \Rightarrow v' = (1-n) y^{-n} \cdot y'$

$\xrightarrow{\boxed{y = v y^n}}$
 $y' = \frac{v' y^n}{(1-n)}$. substitute into equ(1) to obtain:

(2)

$$\frac{1}{1-n} v' + p(x) v = q(x)$$

Step (2) $v' + (1-n) p(x) v = (1-n) q(x)$

Integrating factor method

Ex) Solve the following IVP:

$$x \frac{dy}{dx} + y = xy^2 \xrightarrow{x>0} y(1)=2 \quad [\text{Not in standard form}]$$

$$\boxed{\frac{dy}{dx} + \frac{1}{x} y = y^2}, \quad \boxed{n=2}, \quad \boxed{p(x) = \frac{1}{x}}, \quad \boxed{q(x)=1}$$

Solution ① let $v = y^{1-2} = y^{-1} \Rightarrow v = \frac{1}{y} \Rightarrow \boxed{y = \frac{1}{v}}$

$$\textcircled{2} v' + (1-n) p(x) v = (1-n) q(x)$$

$$v' - \frac{1}{x} v = -1, \quad p(x) = -\frac{1}{x}, \quad q(x) = -1$$

$$\mu(x) = e^{\int -\frac{1}{x} dx}$$

$$= \frac{-\ln(x)}{e} = \frac{\ln(x^{-1})}{e} = \boxed{\frac{1}{x}}$$

$$v(x) = \frac{1}{\frac{1}{x}} \left[\int \frac{1}{x} (-1) dx + c \right]$$

$$v(x) = x [-\ln(x) + c] \Rightarrow \boxed{v(x) = x \ln\left(\frac{1}{x}\right) + cx}$$

$$\text{But } y = \frac{1}{v} \Rightarrow y(x) = \frac{1}{x \ln\left(\frac{1}{x}\right) + cx} \quad (3)$$

$$\text{But } 2 = y(1) = \frac{1}{1 \ln(1) + c(1)} \Rightarrow 2 = \frac{1}{c} \Rightarrow \boxed{c = \frac{1}{2}}$$

$$\boxed{y(x) = \frac{1}{x \ln\left(\frac{1}{x}\right) + \frac{x}{2}}}$$

Ex #2 $\frac{dy}{dx} + \frac{1}{3}y = e^x y^4, \quad y(0) = 1$

$$\boxed{p(x) = \frac{1}{3}} \quad \boxed{q(x) = e^x}$$

Solution standard Bernoulli D.E with $\boxed{n=4}$

$$\textcircled{1} \text{ let } v = y^{1-n} = y^{1-4} \Rightarrow v = y^{-3} \Rightarrow \boxed{v = \frac{1}{y^3}}$$

$$\boxed{y^3 = \frac{1}{v}}$$

$$\textcircled{2} v' + (1-n)p(x)v = (1-n)q(x)$$

$$v' + (1-4)\left(\frac{1}{3}\right)v = -3e^x$$

$$\boxed{v' - v = -3e^x} \quad \text{--- (*)}$$

$$\boxed{p(x) = -1}, \quad \boxed{q(x) = -3e^x}$$

$$M(x) = e^{\int -1 dx} = \boxed{e^{-x}}$$

$$v(x) = \frac{1}{e^{-x}} \left[\int e^{-x} (-3e^x) dx + c \right]$$

$$v(x) = e^x \left[\int -3 dx + c \right]$$

$$v(x) = e^x [-3x + C]$$

$$v(x) = -3xe^x + Ce^x$$

$$\boxed{v(x) = e^x [C - 3x]} \quad \cdot \quad \text{But } y^3 = \frac{1}{v} = \frac{1}{e^x(C-3x)}$$

$$= \frac{e^{-x}}{C-3x}$$

$$\boxed{y(x) = \sqrt[3]{\frac{e^{-x}}{C-3x}}}$$

$$\text{But } 1 = y(0) = \sqrt[3]{\frac{e^{-0}}{C-3(0)}} \Rightarrow 1 = \sqrt[3]{\frac{1}{C}} \Rightarrow \frac{1}{C} = 1 \Rightarrow \boxed{C=1}$$

$$\therefore y(x) = \sqrt[3]{\frac{e^{-x}}{1-3x}}$$

H.w Solve $xy' + y = x^2 y^3$. I.C. $y(1)=1$, $x > 0$

$$y' + \frac{1}{x}y = xy^3$$